

AI-Assisted Performance-Target-Driven Inverse Design for Energy-Efficient Office Facades

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KEYWORDS

Performance-driven Design
Facade Design
Inverse Design
Building energy performance
Surrogate Modelling
Machine Modelling

ABSTRACT

- 1) AI-assisted reverse engineering framework for the early stages of office facade design.
- 2) A machine-learning surrogate model is developed to predict building EUI from different facade design parameters.
- 3) Multiple feasible combinations of WWR, SHGC, U-value, and shading depth are generated to satisfy a predefined EUI target.
- 4) A preliminary EnergyPlus sensitivity study confirms the influence of WWR on the energy performance of the reference office model.

INTRODUCTION

In traditional initial façade design, architects often use the forward design process. They first define the window dimensions and materials, then do energy simulations, and finally get performance results. However, architects often find that it is difficult to choose the right or suitable window-to-wall ratio at this very first stage. A certain window size may meet daylighting requirements, but it may also increase cooling energy use especially in the tropical regions. Traditional methods rely on repeated simulations and multiple comparisons. Essentially, this is a time-consuming process of trial and error, which makes it difficult to systematically explore the design space.

Some regions, such as Singapore, are continually tightening requirements for energy efficiency and low-carbon building design¹. It is increasingly important to predict and manage energy consumption during the operational phase from the outset of the design process. However, most of the tools currently available for simulating building performance operate on the principle of “inputting design data as input and building performance as output.” Designers enter numerous design parameters, and these tools then produce how the building will perform. These tools cannot directly answer the reverse question: 'To

achieve a specific energy consumption target, what facade design parameter combinations should be used?' This limitation highlights the need to develop performance-based predictive modeling methods.

The novelty of this research lies in the use of an inverse performance prediction model. By combining the prediction model with a genetic algorithm, this study aims to develop a design process that shifts the focus from the objective to the parameters. Instead of using simulation tools specifically to study the efficiency of an existing project, this method takes a given energy target as its starting point to generate possible components of a sustainable project. Compared to traditional trial-and-error methods, this approach can generate a range of sustainable design options that meet the energy target, thereby reducing the need for iterative manual adjustments.

The significance of this study can be explained in two ways. From a methodological perspective, this study provides information on prefabrication for the design of energy-efficient buildings. From a practical perspective, it helps architects to take energy efficiency and design requirements into account in the early stages of a project. It can also help to minimise the number of design changes and revisions at later stages.

¹ Inverse design optimization of building EUI targets: generating diverse solutions and enhancing decision interpretability

METHODOLOGY

The study proposes a three-step framework for designing performance-target-driven inverse facade. The framework combines building performance simulation, parametric modelling, machine learning, and optimization. In the first phase, a benchmark model of an office building was built to generate data on the building energy performance of the facade design parameters. In the second phase, machine learning models are trained and compared to select the best model that best classifies the proposed solution. In the third phase, the selected surrogated model options are linked to all best practices. This is used to define a well-defined EUI objective and search for suitable designs, then generate multiple facade parameter combinations which can meet the target.